Background and Introduction

Indonesia is an extremely diverse archipelago comprising 17,508 islands, five main ones, making it the largest in the world. While there is one national government for Indonesia consisting of three branches of government, including an executive branch with a president, there are many differences within each of the islands, which require different governing and regulations. There is even great diversity within one island; for example, Sulawesi is home to around 120 languages and numerous ethnic groups. Indonesia is home to the largest muslim population in the world, which contributes to the governing and social aspects of the country. While it is not a formal Islamic government, historically, only Muslim presidents are elected, and their religious ideals have an influence on the way they govern. This must be kept in mind when considering sustainable finance options, as there are some ways that Islamic ideals can be incorporated, making them more likely to be widely accepted by the large muslim population as well as the government.

Indonesia is the fourth-largest emitter of carbon, particularly due to deforestation (Climate Transparency, 2022). In 2021, the government of Indonesia submitted a strategy that works towards net-zero emissions by 2060, but this strategy was not an official goal (Climate Action Tracker). In 2024, the president declared that they would aim for net-zero by 2050 instead. According to BIOFIN, Indonesia belongs to the 17 mega-diverse countries in the world, and hosts 17 percent of the world's wildlife (BIOFIN). This biodiversity is being degraded due to deforestation, overexploitation of land and resources, tourism, and other human activities. While there are already numerous biodiversity conservation efforts around Indonesia, Indonesia currently faces a USD 1.7 trillion financing gap in order to achieve its sustainable development goals. For biodiversity specifically, the financing gap is estimated to be USD 700 billion. Many "protected areas" exist in Indonesia, but this type of protection requires the least amount of money, and in turn, these areas are often the ones that are least protected due to a lack of enforcement. There are very few protected areas in Indonesia that generate enough income to sustain themselves. These areas are typically funded through international public funding, which clearly does not provide enough support to properly manage these areas. While it is possible that a private donor will provide financial support for a project, the project can typically last only until the funding runs out. This demonstrates the need for sustainable financing options to work towards reaching sustainable development goals, including biodiversity conservation, and have long-term success and results. These strategies include using Islamic finance, payments for ecosystem services, carbon and biodiversity credits, sustainable commodity supply chains, and ecotourism.

Sustainable Financing Background

Numerous sustainable financing options can generate support for biodiversity conservation, but only some can be effective in Indonesia. Many of the mechanisms can seem straightforward in theory, but are more complex upon implementation and management. Some financing schemes require policy from the government to support or allow their implementation. Competing interests between national governments, local governments, and individuals might cause further difficulties in executing the initiatives. For example, the national government is typically focused on short-term generation of wealth and will not see the benefits of some mechanisms that generate wealth in the long run. As Birdlife and Burung Indonesia state in their

project document regarding conservation efforts in Mbelling, sustainable finance mechanisms "must generate continuous revenue streams, create strong local incentives for conservation and restoration, and embed green development as a competitive economic pathway".

Islamic Financing

Green sukuk is a mechanism that complies with Islamic law in order to finance green initiatives. Sukuk is a certificate ensuring a financial bond follows Islamic, or Sharia law, meaning it prohibits interest-based transactions. With typical bonds, the owner receives interest payments, while the owner of a sukuk becomes a partial owner of an asset that generates income (Positive Income). For green sukuk, 100% of the finance goes towards "green" projects, including mitigation and adaptation to climate change, as well as biodiversity conservation (Indonesia Ministry of Finance). In 2018, the United Nations Development Program (UNDP) and the Innovating Financing Lab published Indonesia's Green Sukuk Initiative, in which UNDP outlines its support for the government regarding sustainable financing. They outline 5 main sectors that the project will focus on: renewable energy, sustainable transportation, resilience to climate change and disaster risk areas, waste-to-energy and waste management, and sustainable agriculture (UNDP).

The Government of Indonesia, through the Ministry of Finance, established its first sovereign green sukuk in 2019 with an amount of 1.25 billion USD. This sukuk will go to 'eligible green projects', which are determined by the Green Bond and Green Sukuk Framework. In 2025, Indonesia issued around 9.59 billion USD in green sukuk cumulatively since 2019 (Ministry of Finance, Republic of Indonesia, 2024). The money for this sukuk comes from investors, both domestic and international, but so far, most funds seem to be coming from international sources. In the yearly performance reports published by the Ministry of Finance Republic of Indonesia, they outline projects that the funds go to each year. In 2024, these included planning, development, and oversight of renewable energy and energy conservation Infrastructure, provision of geological data, information, recommendations, and services, and planning and oversight of watershed management, development of forest seedling production, and forest rehabilitation. All projects to date are owned by government agencies like the Ministry of Environment and Forestry. Read Performance Reports from the Ministry of Finance, Republic of Indonesia, to see a detailed yearly breakdown of green sukuk from 2019 to 2024. I see great potential for green sukuk to be used towards conservation efforts that can generate income and benefits. For example, the government could use this money to invest in smallholder farming practices and agroforestry, which will decrease deforestation and later bring income from crop production. Although the allocation of these funds is labelled as "green", the government retains control and may not allocate the funds to the most necessary places.

Islamic finance can come from other sources to contribute to sustainable goals. Zakat is a financial contribution of 2.5% that Muslims are obliged to pay if they earn a certain amount. Typically, this contribution is used towards poverty alleviation, but can also be used towards environmental protection and climate efforts. In Indonesia, the Zakat potential is estimated to be around USD 21 billion (UNDP). In March of this year, UNDP Indonesia convened with Badan Amil Zakat Nasional (BAZNAS) and Bank Syariah Indonesia (BSI) to develop a green Zakat framework. They discussed how these funds can "contribute to climate action and social resilience in an inclusive and transformative way, adding its value and expanding its impact to

local communities" (UNDP). BSI contributes 2.5% of its income to corporate Zakat, and has recently launched a green Zakat initiative, unlocking a good sum of money to be contributed towards environmental protection initiatives.

Islamic finance can be an effective form of sustainable financing in Indonesia because it appeals to the predominantly Muslim population and government while also working towards the Sustainable Development Goals. One current drawback of this method is that the national government is in control of the funds, and is too far from the ground to know the actual needs. Smaller entities can be left out of the conversation, and their input may not be considered. This could be remedied by having representatives from NGOs as well as individuals from local or indigenous communities present when determining the allocation of these funds from both green Sukuk and green Zakat. These are large sources of funding that could go towards beginning other sustainable finance initiatives that require support to get started, and work towards making money to sustain themselves.

Payments for Ecosystem Services

Payments for ecosystem services, as the name suggests, are payments for services like conserved or restored forests, improved sustainable livelihoods for forests, communities, preserved biodiversity, and reduced carbon emissions. Through this scheme, those who are benefiting from these services should be paying for their provision, which is true of most other goods and services.

An example of this comes from the sustainable financing plan of Birdlife and Burung Indonesia in Mbelling. Their payment for ecosystem services scheme involves encouraging water bottle companies and retailers to contribute to watershed conservation. Currently, the small-scale water bottle company, Amandava, is only making enough to cover the salaries. In the future, the hope is that the profit from the sale of water bottles will be used towards the conservation of the watershed supplying the water.

Through a study of all PES schemes in Indonesia, Suich et al. found benefits and constraints regarding the method. In this study, only payment schemes for water and carbon were analysed. Benefits were clear, that PES schemes deliver social and economic benefits to communities while also protecting natural environments. Ecosystem services can be clearly defined and measured, and payments will be made only when they are satisfied. Constraints included "a lack of recognition that degradation and scarcity of ecosystem services is a problem, and the constraints imposed by the conflicting and uncertain regulatory environment" (Suich et al., 2016).

Partnership for Forests (P4F) published a case study of two PES projects in Indonesia. Both projects are located in Kapuas Hulu District, West Kalimantan, specifically the Nanga Lauk village forest and Tanjung. P4F outlines 6 steps in any PES value chain. First, assessment and identification of market demand, then finding and getting supply projects ready (feasibility, project design, budgeting), next, long-term purchase service agreement with PES buyers, then due diligence, certification process for the supply chain project. Finally, implementation, annual monitoring and reporting, and claiming of conservation outcomes by buyers and payment of suppliers (P4F, 2018). In Nanga Lauk and Tanjung, an asset management company called Lestari Capital manages Sustainable Commodities Conservation Mechanism (SCCM) and the Rimba Collective. SCCM and Rimba collective finance projects over a 25-year period, and tie

payments to results on the ground. For example, they provide financial and technical assistance for companies to become Roundtable on Sustainable Palm Oil (RSPO) certified. Through Rimba collective and SCCM, companies are able to buy certified conservation outcomes from village forests and ecosystem restoration concessions (ERCs), which can improve their footprint and help offset to meet RSPO standards. These private companies may invest to compensate for past deforestation and meet sustainable palm oil goals. In turn, village communities gain the resources and capacity to sustainably manage their forests, which they are often unable to do without this added support from private funding. This initiative was launched in 2018, and there are few updates about its current success. Since this is over a 25-year period, it will be a long time before we know the overall success of the project, but it seems to be a potentially effective scheme that could be replicated in other areas. The focus on long-term sustainability and community empowerment is an indicator that they are moving in the right direction. PES mechanisms can be very effective as they are often long-term efforts, and funding is indefinite. As long as the service continues, the payment can continue.

Biodiversity and Carbon Credits

In 2023, Indonesia launched Indonesia Carbon Exchange (IDX Carbon/Bursa Karbon Indonesia), which is a platform for Indonesian carbon trading. With this platform, carbon can be traded as credits, with one unit being 1 ton of CO_2 equivalent (tCO_2e) reduced or removed from the atmosphere. Companies with emission caps can buy more credits to allow them to emit more, and companies without caps can buy credits to offset their emissions. It is also possible that companies actively removing carbon from the atmosphere can get credits and sell them to generate revenue. State-owned and private companies, financial institutions, and developers can all trade on this platform, but they must be registered with IDXCarbon. Current projects available for trading in Indonesia are all within the energy sector, but forestry projects are possible.

One forestry project with carbon credits is REDD+, Reducing Emissions from Deforestation and Forest Degradation plus Conservation, Sustainable forest management, and Enhancement of forest carbon stocks. Through this, groups can be paid for restoring or protecting forests based on the amount of carbon that was saved, through credits. These credits do not have to be sold through IDXCarbon and can also be sold through voluntary markets. There are still conditions to trade outside of IDXCarbon, but a carbon project is able to sell credits directly to a buyer if it follows them. Often, these credits are generated on ‘carbon concessions’, areas established for this purpose.

The Katingan REDD+ Project in Indonesia is a carbon concession area where carbon trading occurs, but it can be seen as a cautionary tale of the potential damage that can be caused by this method. In 2013, this area was granted an ERC license to carry out peatland restoration. There were local communities located outside of this concession area, but they were considered part of the “project area” and were displaced and forced to change their livelihoods to comply with REDD+ guidelines. For example, the Bapinang Hilir community has numerous small farmers who have their practices, including controlled burning before planting their crops to prevent the need for herbicides. After the REDD+ project began in the area, the farmers were required to stop practices like these, and continuing them could be punishable by jail, forcing them to start using herbicides. In the meantime, carbon credits were being sold to

companies like Shell and KLM, giving them the ability to claim they are ‘carbon neutral’. Additionally, deforestation is still happening in this area, demonstrating that carbon credits aren’t always what they seem or claim to be.

While the carbon credit scheme can be beneficial to generate revenue and decrease carbon emissions, the current state of the project in Indonesia has much room for improvement. Since this initiative is so new, there is a need for much development and growth. Monitoring, reporting, and verification are required to ensure companies are being true to their word.

Biodiversity credits are similar to carbon credits, but are far more difficult to define. The biodiversity finance initiative (BIOFIN) and UNDP explain, “Biodiversity credits aim to capture the value of nature, not for its extractable resources, but for its integrity, richness, and resilience” (BIOFIN, 2025). The Biodiversity Credit Alliance explains that biodiversity credits represent verified positive outcomes for biodiversity, including the restoration of forests, protection from potential harm, and maintaining intact forests. These outcomes can be much harder to identify and define, making them more nuanced than carbon credits, although they can be bought and traded in the same way. Through this pilot program, BIOFIN identified five main challenges to scaling biodiversity credits. These included policy and legal gaps, methodological ambiguity, market skepticism, high cost of implementation, and institutional capacity (BIOFIN, 2025). Indonesia has no policy regarding biodiversity credits, and as of now, it must rely on the precedent set for carbon credit guidelines. For this program to be scaled up, there must be clear legality and governmental guidance regarding biodiversity credits.

Indonesia is piloting a program with biodiversity credits in Harapan Forest, the last remaining lowland tropical rainforest in Sumatra. In this forest, a private company (PT REKI) was granted an Ecosystem Restoration Concession (ERC) license by the Indonesian government. This license allows them to carry out forest restoration on ca. 100,000 ha of land. This area was chosen as it has become a forest island surrounded by harmful agricultural lands. The success of these ERC areas is unclear. Credits could be a way for funding to continue and pay for the implementation and monitoring of conservation efforts. This is the first area with this type of concession, which could be a great model for other areas to follow.

Carbon and biodiversity credits have the potential to generate revenue that can be used towards biodiversity conservation, but a legal framework must be in place for a project to be set up and run. In many important landscapes in Indonesia, the land is owned by many different people. This includes different community members, companies, and the government with concessions. This distribution of ownership makes it difficult to create policy in the whole area, which applies to one for a credit scheme.

Another concern about both types of credits is their involvement in greenwashing. Companies that emit a lot of carbon or contribute to biodiversity loss can buy credits to offset this damage, and claim they are being “sustainable” or benefit the environment. This is misleading as they are not actually being sustainable, and are causing much harm. Most consumers do not know this, and may think they are supporting a carbon-neutral business that is causing no environmental damage. The main remedy to avoid greenwashing is full transparency from the companies and scientific rigor to track emissions and sustainability.

As with any project, the most important thing is that local communities are included in the conversation. Their insight and experience are invaluable to creating a project that will be successful and sustainable for everyone involved.

Sustainable Commodity Supply Chains

In the last 20 years, farmers have already transitioned from shifting agriculture, which requires deforestation, to agroforestry, growing commodities like candlenut, which is used in many beauty products.

An example of sustainable supply chains is Bird Life's "living rubber project" with the Indigenous People launched in 2021. This project was in partnership with BMW Group and Pirelli and aimed to protect the Hutan Harapan rainforest in Sumatra. This project supports the sustainable production of natural rubber while protecting the people, animals, and plants in the area. To do this, community members will be provided with training on how to continue their farming while supporting the environment. Indigenous people, particularly the Batin Sembilan who are native to this rainforest, are deeply intertwined in the process and have a say about how this habitat is managed. This tire and car company is working with an NGO to sustainably source their rubber, and has a traceable supply chain that also provides income to local communities. To do this, they will train in an agr-forestry approach which "will allow growers to optimise the relationship between production, soil fertility, diversification of revenue streams (there will be different plantings in the same area with a view to reducing small-holders' dependence on just natural rubber), conservation of biodiversity and capitalisation of resources in a circular way" (BMW Group). This is a great example of sustainable commodity supply chains that were effective and can be replicated in other areas of Indonesia. If companies at the end of the supply chain, the ones creating the product, provide initial funding to transition to sustainable practices, then the local community can adopt these and start making a profit for salaries and to contribute to biodiversity conservation in their areas.

In 2023, the EU passed a regulation called the European Union Deforestation Regulation (EUDR) specifying that specific commodities (cattle, cocoa, coffee, oil palm, rubber, soya, and wood) cannot be sold in the EU if they have been involved in deforestation after the year 2020. This requires companies to have completely traceable supply chains, giving consumers a choice to buy sustainably. The EU is a major economy and consumer of these products, and its regulations will force supply chains around the world to rethink their agricultural practices. The issue is that some farmers may not have the resources and knowledge to completely change their farming practices and require support to begin this shift. It must be considered the possibility of a completely sustainable supply chain that is not doing anything for biodiversity. It is important, and challenging, that the financing mechanism is sustainable, generates an income supporting the people, and also protects biodiversity.

An innovation to aid with this knowledge gap is an app called Sawit Kita, which provides farmers with courses to learn Good Agriculture Practices or GAP and Best Management Practices or BMP for sustainable palm oil management. This was presented in 2022 at the Securing Sustainability in Turbulent Time – Sustainable Palm Oil Dialogue event to discuss the accelerating impact of sustainable palm oil. Innovations like these can be used in conjunction with financial support to create supply chains that can be developed to sustain themselves. This kind of innovation can be beneficial for communities that have access to phones and the internet, which is not true of many, making it necessary for training in accessible ways.

Ecotourism

Another way to generate income in a way that benefits different parties is through ecotourism. Ecotourism is a responsible way to travel, respecting nature and local people while supporting the local economy. The issue is that there is no proper definition of the term, and it is not something that is easily regulated.

An example of an ecotourism project is in Komodo National Park, a UNESCO Heritage Site. This park is made up of three large islands between Sumbawa and Flores, along with smaller surrounding islands all within Wallacea and including parts of the Mbelling landscape. The Komodo dragon is endemic to this area, and can be found in abundance, making it a desirable destination for tourists from around the world. Some consider this ecotourism as it uses the endemic species and natural landscape to attract tourists, and the entrance fees and tour operations help fund conservation in the area. This national park is managed by the central government of Indonesia through the Directorate General of Forest Protection and Natural Conservation of the Ministry of Forestry. Since it is managed by the government, the decisions made regarding it have motives other than conservation. The Indonesian government has identified this area, specifically the port town of Labuan Bajo, as a potential mega tourist destination and a “new Bali”. With the determination to increase economic growth, the Ministry of Environment and Forestry has begun changing previously protected areas to “utilization areas” and allowing private investment. This will lead to the development of resorts and activities that will be harmful to the environment and species, and most of the proceeds will not reach local communities. This is no longer ecotourism, and it begins to cause more harm than good to locals, plants, and animals in the area.

The issue with ecotourism is that it is hard to enforce, and there is a very thin line between ecotourism and tourism that is destructive and harmful. Two examples of ecotourism that are more effective at protecting the environment are Bwindi Impenetrable National Park and Volcanoes National Park. At both of these sites, permits are required to visit, and only a certain number of permits are granted at a higher price. This means that the number of visitors in the park is limited, decreasing habitat degradation, and there is more income from the permits that can go to local communities. It must be noted that this creates an unjust system in which only people with the financial means can enjoy this nature.

While, when done right, ecotourism can lead to sustainable sources of income as well as biodiversity protection, it is so hard to ensure that ecotourism is not damaging the environment. All kinds of tourism, including ecotourism, increase the need for water, agriculture, and energy to support the larger number of people in the area. At the moment, with a lack of resources and funding, it seems important to prioritize other forms of protection before putting time and energy into ecotourism schemes.

Recommendations

No one financing mechanism will be sufficient in any area of Indonesia or the world. A combination of these mechanisms must be used in areas, dependent on the landscape, local

communities, and policies. Any project must consider long-term outcomes and be a plan over 30-50 years. Additionally, education is always an answer and can be an important first step towards expanding sustainable financing schemes across Indonesia. If people don’t know about the damage to their environment, they will have no incentive to pay for or start incentives to protect it. Additionally, the barrier often comes when communities do not have access to training and information on how to change their practices to protect biodiversity.

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